concurrency_deadlock_detection (Initial Draft)

Domain: Operating Systems | Document ID: concurrency_deadlock_detection_v1
Specification Standard: IEEE Capstone Evaluation Corpus

Executive Overview:

Coffman conditions, resource allocation graphs, and Banker’s algorithm
for deadlock avoidance in multi-threaded systems.

Methodological Verification & System Lineage:

This specification is catalogued in IntentCloud for intent-aware hybrid retrieval.
Text chunks are indexed into 384-dimensional dense semantic vectors and
1,000,003-dimensional sparse feature-hashed token projections.
Candidate pools are unified using scale-invariant Reciprocal Rank Fusion (k=60).
Explainable citation attribution is extracted using cross-encoder cross-attention.